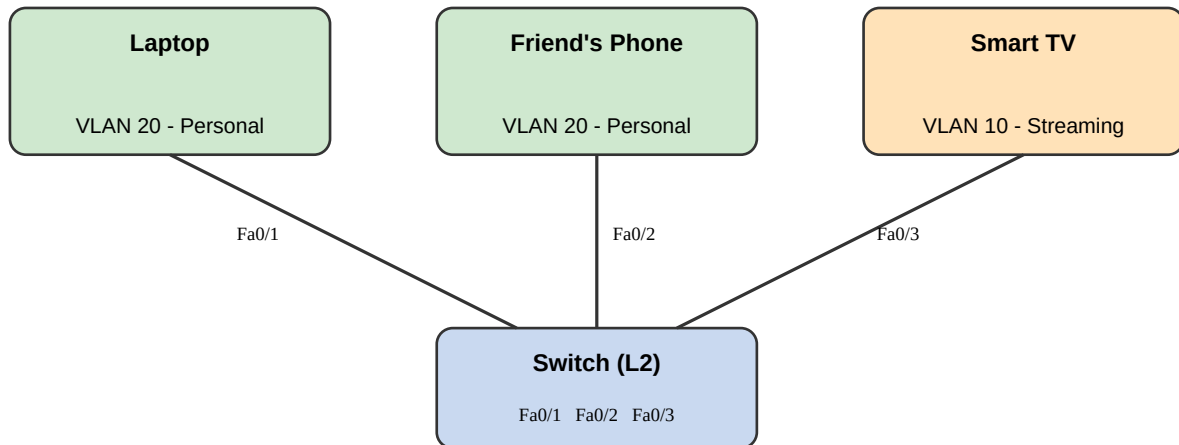


Switching and VLANs - Solutions

Four practical exercises with diagram, Cisco IOS configuration (Packet Tracer / NetSim syntax), verification and automation.

Question 1: Network diagram with VLAN labels

Draw a simple network diagram showing three devices (laptop, friend's phone, smart TV) connected to a switch, and label which device is in which VLAN.



VLAN 20 (Personal)

VLAN 10 (Streaming)

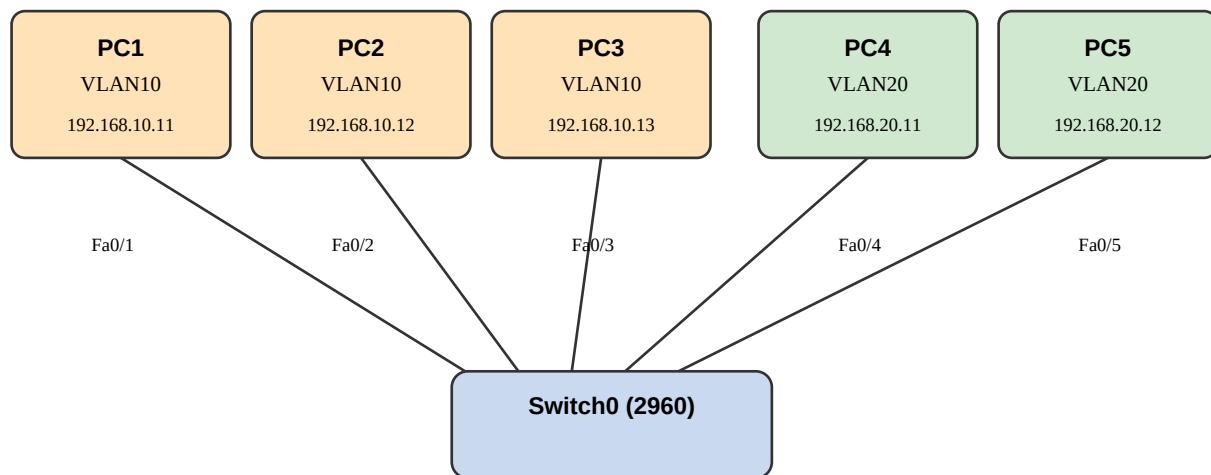
Device	Switch port	VLAN	Reason
Laptop	Fa0/1	VLAN 20 (Personal)	Private data stays on personal segment
Friend's phone	Fa0/2	VLAN 20 (Personal)	Personal device, same trust level
Smart TV	Fa0/3	VLAN 10 (Streaming)	Less trusted IoT/streaming traffic isolated

Because the TV is in a different VLAN, it is a separate broadcast domain. A compromised or noisy streaming device cannot directly reach the laptop or phone at Layer 2.

Question 2: Create VLAN 10 and VLAN 20, assign PCs, verify isolation

Create two VLANs, assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify PCs in different VLANs cannot ping each other.

Topology and addressing



PC	Port	VLAN	IP address	Subnet mask
PC1	Fa0/1	10	192.168.10.11	255.255.255.0
PC2	Fa0/2	10	192.168.10.12	255.255.255.0
PC3	Fa0/3	10	192.168.10.13	255.255.255.0
PC4	Fa0/4	20	192.168.20.11	255.255.255.0
PC5	Fa0/5	20	192.168.20.12	255.255.255.0

Switch configuration

```

Switch> enable
Switch# configure terminal
Switch(config)# hostname SW1

! --- Create VLANs ---
SW1(config)# vlan 10
SW1(config-vlan)# name SALES
SW1(config-vlan)# exit
SW1(config)# vlan 20
SW1(config-vlan)# name HR
SW1(config-vlan)# exit

! --- Assign ports to VLAN 10 (PC1-PC3) ---
SW1(config)# interface range fastEthernet 0/1 - 3
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 10
SW1(config-if-range)# exit

! --- Assign ports to VLAN 20 (PC4-PC5) ---
SW1(config)# interface range fastEthernet 0/4 - 5
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 20
SW1(config-if-range)# end
SW1# write memory

```

Verification

```
SW1# show vlan brief
```

```

VLAN Name      Status   Ports
-----

```

```

1    default    active    Fa0/6 ... Fa0/24, Gi0/1, Gi0/2
10   SALES      active    Fa0/1, Fa0/2, Fa0/3
20   HR         active    Fa0/4, Fa0/5

```

Test (from PC command prompt)	Expected result	Why
PC1> ping 192.168.10.12 (PC2)	Success (0% loss)	Same VLAN 10
PC1> ping 192.168.10.13 (PC3)	Success (0% loss)	Same VLAN 10
PC4> ping 192.168.20.12 (PC5)	Success (0% loss)	Same VLAN 20
PC1> ping 192.168.20.11 (PC4)	Request timed out / Destination host unreachable	Different VLAN, no router
PC3> ping 192.168.20.12 (PC5)	Request timed out / Destination host unreachable	Different VLAN, no router

Conclusion: VLANs split the switch into separate broadcast domains. Without a router (inter-VLAN routing), hosts in VLAN 10 cannot reach VLAN 20 - even though they plug into the same physical switch. Note: the first ping may time out because of ARP; check the remaining replies.

Question 3: Move a PC from VLAN 10 to VLAN 20

Move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity.

Scenario: PC3 (port Fa0/3) is moved from VLAN 10 to VLAN 20 using the same topology as Question 2.

Before

```

SW1# show running-config interface fastEthernet 0/3
interface FastEthernet0/3
    switchport access vlan 10
    switchport mode access

SW1# show vlan brief
10   SALES      active    Fa0/1, Fa0/2, Fa0/3
20   HR         active    Fa0/4, Fa0/5

PC3> ping 192.168.10.11    -> Success (same VLAN as PC1)
PC3> ping 192.168.20.11    -> Fails (PC4 is in VLAN 20)

```

Change commands

```

SW1# configure terminal
SW1(config)# interface fastEthernet 0/3
SW1(config-if)# switchport access vlan 20
SW1(config-if)# end
SW1# write memory

```

Then update PC3's IP to the VLAN 20 subnet (Desktop > IP Configuration): IP 192.168.20.13, mask 255.255.255.0. Without this, PC3 would be in the VLAN 20 broadcast domain but with a VLAN 10 address, and pings would still fail.

After

```

SW1# show vlan brief
10   SALES      active    Fa0/1, Fa0/2
20   HR         active    Fa0/3, Fa0/4, Fa0/5

PC3> ping 192.168.20.11    -> Success (now same VLAN as PC4)
PC3> ping 192.168.20.12    -> Success

```

PC3> ping 192.168.10.11 -> Fails (PC1 stays in VLAN 10)

Test	Before	After
PC3 to PC1 (VLAN 10)	Success	Fail
PC3 to PC4 (VLAN 20)	Fail	Success
PC3 to PC5 (VLAN 20)	Fail	Success

Question 4: Automate assigning 10 ports to VLAN 30

Write a short script or command list that assigns 10 ports on a switch to VLAN 30, using a loop or range command.

Option A: Cisco IOS range command (Packet Tracer / NetSim / GNS3)

```
SW1# configure terminal
SW1(config)# vlan 30
SW1(config-vlan)# name SERVERS
SW1(config-vlan)# exit
SW1(config)# interface range fastEthernet 0/1 - 10
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 30
SW1(config-if-range)# no shutdown
SW1(config-if-range)# end
SW1# write memory
SW1# show vlan id 30
```

The single command `interface range fastEthernet 0/1 - 10` applies the next commands to ports Fa0/1 through Fa0/10, so nothing is repeated 10 times. Non-contiguous ports can be listed with commas, e.g. `interface range fa0/1 - 5 , fa0/8 - 12`.

Option B: Python loop (works with GNS3 or any switch reachable over SSH/telnet)

```
# Requires: pip install netmiko
from netmiko import ConnectHandler

switch = {
    "device_type": "cisco_ios",
    "host": "192.168.1.2",
    "username": "admin",
    "password": "cisco123",
    "secret": "cisco123",
}

commands = ["vlan 30", "name SERVERS"]
for port in range(1, 11):
    commands += [f"interface fastEthernet 0/{port}",
                  "switchport mode access",
                  "switchport access vlan 30"]

conn = ConnectHandler(**switch)
conn.enable()
print(conn.send_config_set(commands))
print(conn.send_command("show vlan id 30"))
conn.save_config()
conn.disconnect()
```

Expected verification output:

VLAN Name	Status	Ports
-----------	--------	-------

```
30  SERVERS    active  Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5,  
                                Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10
```

Note: Port names differ by switch model (Fa0/x on 2960, Gi1/0/x on 3750/3850). Adjust the range accordingly.